

Solving Inequalities with the Distributive PropertyWorksheet 7A.01.09 **ANSWER KEY**

For questions 1 through 4, use the inequality $-30 \geq -6(8 - x)$.

1. Solve it by using the distributive property first.

$$-30 \geq -48 + 6x \quad (\text{distribute the } -6 \text{ to both terms})$$

$$18 \geq 6x \quad (\text{add } 48 \text{ to both sides})$$

$$3 \geq x, \text{ so } x \leq 3 \quad (\text{divide both sides by } 6 \text{ — no reversal})$$

Distributing a negative factor is not the same as dividing by one. Nothing reverses here, because the only division is by a positive 6.

2. Solve it again by eliminating the -6 first.

$$5 \leq 8 - x \quad (\text{divide both sides by } -6 \text{ and REVERSE})$$

$$-3 \leq -x \quad (\text{subtract } 8 \text{ from both sides})$$

$$3 \geq x, \text{ so } x \leq 3 \quad (\text{multiply both sides by } -1 \text{ and REVERSE again})$$

Two reversals this way and none the other way, and both routes land on $x \leq 3$. The number of flips is not what makes an answer right.

3. Graph the solution on the number line.



Closed circle at 3, shaded left. The symbol is $\geq$, so the boundary is included.

4. Verify your solution set by substitution. Test one value inside it and one value outside it.

$$\text{Inside: } x = 0 \rightarrow -6(8 - 0) = -48, \text{ and } -30 \geq -48 \checkmark$$

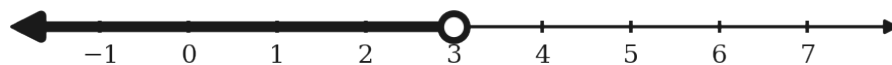
$$\text{Outside: } x = 5 \rightarrow -6(8 - 5) = -18, \text{ and } -30 \geq -18 \times$$

$$\text{Boundary: } x = 3 \rightarrow -6(8 - 3) = -30, \text{ and } -30 \geq -30 \checkmark$$

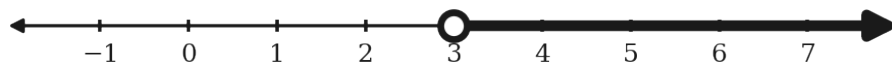
An inside value on its own is not enough: a solution set that is too narrow passes it every time, because everything inside a too-narrow answer really is a solution. The outside value is what catches that. A reversed symbol shows up differently — the two results SWAP, inside false and outside true.

5. Which graph represents the solution to $-(a + 4) > -7$?

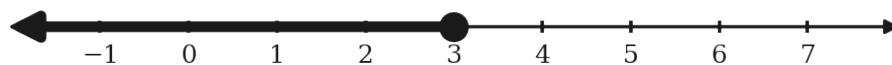
A. **correct**



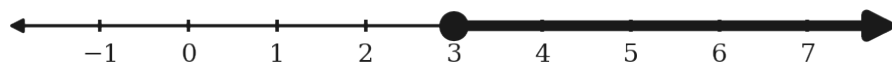
B.



C.



D.



$-(a + 4) > -7$ is $-a - 4 > -7$, so $-a > -3$, so $a < 3$.

Choice A. B has the direction reversed, C and D use a closed circle for a strict symbol. Dividing by -1 at the last step is the reversal students skip, and skipping it produces choice B exactly.

For questions 6 through 9, solve using whichever method you prefer, then graph the solution set.

6. $9 > 3(5c - 7)$

$3 > 5c - 7$ (divide both sides by 3 — no reversal)

$10 > 5c$

$2 > c$, so $c < 2$

Check: $c = 0$ gives $3(-7) = -21$ and $9 > -21$ ✓; $c = 3$ gives $3(8) = 24$ and $9 > 24$ ✗

Solution. $c < 2$



Dividing by 3 first is cleaner than distributing, which would put $15c - 21$ on the right.

7. $\frac{1}{3}(6 - m) \geq -4$

$6 - m \geq -12$ (multiply both sides by 3 — no reversal)

$-m \geq -18$

$m \leq 18$ (multiply both sides by -1 and REVERSE)

Check: $m = 0$ gives $\frac{1}{3}(6) = 2$ and $2 \geq -4$ ✓; $m = 30$ gives $\frac{1}{3}(-24) = -8$ and $-8 \geq -4$ ✗

Solution. $m \leq 18$



Multiplying by 3 does not reverse anything. The reversal comes later, at $-m \geq -18$.

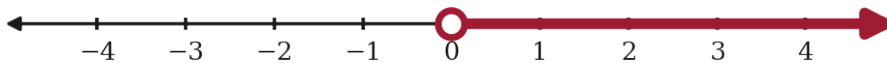
8. $-0.9(x + 2.5) < -2.25$

$x + 2.5 > 2.5$ (divide both sides by -0.9 and REVERSE)

$x > 0$

Check: $x = 1$ gives $-0.9(3.5) = -3.15$ and $-3.15 < -2.25$ ✓; $x = -1$ gives $-0.9(1.5) = -1.35$ and $-1.35 < -2.25$ ✗

Solution. $x > 0$



Dividing by -0.9 reverses the symbol once. Distributing first gives $-0.9x - 2.25 < -2.25$, which works too and still reverses at the last step.

9. $(5/3)(11 - x/8) \geq 20$

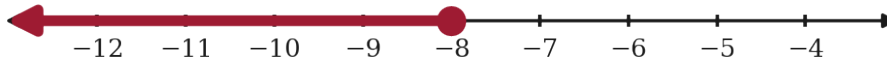
$11 - x/8 \geq 12$ (multiply both sides by $3/5$ — no reversal)

$-x/8 \geq 1$

$x \leq -8$ (multiply both sides by -8 and REVERSE)

Check: $x = -8$ gives $(5/3)(12) = 20$ and $20 \geq 20$ ✓; $x = 0$ gives $(5/3)(11) \approx 18.3$ and $18.3 \geq 20$ ✗

Solution. $x \leq -8$



$20 \div (5/3) = 20 \times (3/5) = 12$. Distributing here is a mess; this is the item that makes the case for eliminating the factor first.

10. A classmate was absent today. Describe, in words, two different ways to solve $2(-7 + d) \leq -16$. Then give the solution.

Way 1 — distribute first. $2(-7) + 2(d) = -14 + 2d$, so $-14 + 2d \leq -16$. Add 14 to both sides: $2d \leq -2$. Divide both sides by 2: $d \leq -1$.

Way 2 — divide first. Divide both sides by 2 to get $-7 + d \leq -8$. Add 7 to both sides: $d \leq -1$.

Solution: $d \leq -1$

Neither way reverses the symbol, because 2 is positive. Accept any clear description; the two methods have to be genuinely different, not the same steps renamed.